

SWIGGY SALES ANALYSIS

SQL Data Analytics Project Report

Project Type: SQL Data Analytics Project

Domain: Food Delivery / E-commerce

Author: Adrij Das

Dataset Size: 1,97,401 records

Data Cleaning & Validation • Dimensional Modelling • ETL • KPI Development
• Business Analysis

Project Documentation

1. Executive Summary

The Swiggy Sales Analysis project is an end-to-end SQL data analytics project developed to analyze food-delivery order data and generate meaningful business insights.

The project works with 1,97,401 Swiggy order records containing information about states, cities, locations, restaurants, food categories, dishes, prices, customer ratings, and rating counts. The raw data was examined for null values, blank strings, and duplicate records before being transformed into a structured Star Schema.

The analytical model was then used to calculate core KPIs such as total orders, total revenue, average dish price, and average customer rating. Further SQL analysis covers monthly, quarterly, yearly, and day-of-week trends, geographic performance, restaurant performance, food-category performance, customer spending patterns, and rating distributions.

Overall workflow: Data Ingestion → Data Quality → Data Cleaning → Dimensional Modelling → ETL → KPI Development → Business Analysis.

2. Project Background

Food-delivery platforms generate large volumes of transactional data involving customers, restaurants, dishes, locations, prices, and ratings. Analyzing this data can help understand order demand, revenue performance, geographic trends, restaurant performance, food preferences, customer spending behavior, customer satisfaction, and potential growth opportunities.

This project uses Swiggy order data to demonstrate how SQL can transform raw transactional data into a structured analytical model and derive business-oriented insights.

3. Problem Statement

The raw Swiggy dataset contains a large number of food-delivery records in a source structure. Direct analysis can be affected by missing values, blank values, duplicate records, repeated descriptive information, and the lack of a dedicated analytical model.

The project therefore focuses on cleaning and validating the raw dataset, creating a dimensional model, loading the cleaned data into the analytical schema, and using SQL to answer important business questions.

4. Project Objectives

- Load and examine the raw Swiggy order dataset.
- Perform comprehensive data-quality checks.
- Identify null and empty-string values.
- Detect duplicate records.
- Remove duplicate records using SQL window functions.
- Design a Star Schema for analytical reporting.
- Create dimension and fact tables.
- Populate the dimensional model using SQL-based ETL.
- Calculate important business KPIs.
- Perform time-based sales analysis.

- Analyze city and state-level performance.
- Analyze restaurant and food-category performance.
- Analyze customer spending behavior.
- Analyze customer rating patterns.
- Identify potential business opportunities from the data.

5. Dataset Description

The project uses a Swiggy food-delivery dataset containing approximately 1,97,401 records.

Column	Description
State	State associated with the order
City	City where the order was placed
Order_Date	Date of the order
Restaurant_Name	Name of the restaurant
Location	Specific location or area
Category	Food or cuisine category
Dish_Name	Name of the ordered dish
Price_INR	Price of the dish in Indian Rupees
Rating	Customer rating
Rating_Count	Number of ratings received

6. Project Methodology

```

Raw Swiggy Data
↓
Data Ingestion
↓
Data Quality Validation
↓
Data Cleaning
↓
Dimensional Modelling
↓
ETL / Data Loading
↓
KPI Development
↓
Business Analysis
↓
Business Insights

```

7. Data Ingestion

The first stage involved loading the raw Swiggy order dataset into the source table `swiggy_data`. The dataset was reviewed to understand its structure and available fields before beginning data-quality validation.

8. Data Quality Assessment

8.1 Null Value Check

SQL queries were used to identify missing values in important business columns, including State, City, Order Date, Restaurant Name, Location, Category, Dish Name, Price, Rating, and Rating Count.

8.2 Empty String Check

The dataset was checked for blank or empty-string values because these can create inconsistent grouping, filtering, and reporting results.

8.3 Duplicate Detection

Duplicate records were identified by grouping across relevant business-critical columns and checking for groups with more than one occurrence.

```
GROUP BY ...  
HAVING COUNT(*) > 1
```

9. Data Cleaning

After identifying duplicate records, a Common Table Expression and the ROW_NUMBER() window function were used to rank duplicate records. One record was retained while surplus duplicates were removed.

```
ROW_NUMBER() OVER (  
PARTITION BY business_columns  
ORDER BY ...  
)
```

This controlled approach prevents duplicate records from inflating order counts, revenue calculations, and other business metrics.

10. Dimensional Modelling

After cleaning the raw data, the project implemented a Star Schema. Descriptive information was separated into dimension tables while measurable business data was maintained in a central fact table.

This structure reduces data duplication, improves query clarity, supports reporting, and provides a scalable foundation for analytics.

11. Dimension Tables

Dimension Table	Key Attributes
dim_date	Year, Month, Quarter, Week
dim_location	State, City, Location
dim_restaurant	Restaurant_Name
dim_category	Cuisine / Category
dim_dish	Dish_Name

12. Fact Table

The central fact table created for the project is fact_swiggy_orders. It stores measurable business information and links to the dimension tables through foreign keys.

Measures: Price_INR, Rating, Rating_Count

Foreign-key relationships: Date, Location, Restaurant, Category, and Dish dimensions.

13. Star Schema Architecture

```
dim_date
|
|
dim_location ---- fact_swiggy_orders ---- dim_restaurant
|
|
dim_category
|
|
dim_dish
```

14. ETL Process

Dimension Loading

Distinct records were extracted from the source dataset and inserted into the corresponding dimension tables.

Fact Loading

The source data was joined with the dimension tables to resolve the required foreign keys before inserting records into the fact table.

The SQL workflow therefore separates source profiling, cleaning, dimension creation, dimension loading, and fact loading into distinct analytical steps.

15. KPI Development

15.1 Total Orders

Measures the total number of orders in the analytical dataset.

```
SELECT COUNT(*) AS Total_Orders
FROM fact_swiggy_orders;
```

15.2 Total Revenue

Calculates total revenue by summing Price_INR and presenting the result in INR millions.

15.3 Average Dish Price

Calculates the average price of dishes across the dataset.

15.4 Average Rating

Calculates the average customer rating.

16. Time-Based Analysis

- Monthly order trends
- Monthly revenue
- Quarterly order trends
- Yearly order volume
- Day-of-week order patterns

- Highest-revenue month
- Month-over-month revenue growth

The analysis uses aggregation, date functions, CTEs, and the LAG() window function for period-over-period comparisons.

17. Geographic Analysis

- Top 10 cities by order volume
- Revenue contribution by state
- Cities with the highest average restaurant ratings
- Percentage of total revenue contributed by each state
- Restaurants that should be prioritized for promotion
- Identification of high-value restaurants

18. Restaurant Performance Analysis

- Restaurants ranked by order volume
- Restaurants ranked by revenue
- Average restaurant rating
- High-value restaurants
- Restaurant prioritization
- Restaurants contributing the highest revenue

The project also combines multiple metrics such as orders, ratings, price, and revenue to prioritize restaurants.

19. Food & Cuisine Analysis

- Top food categories by order volume
- Most ordered dishes
- Cuisine performance using order volume and average rating
- Restaurants and food segments contributing significantly to revenue

20. Customer Spending Analysis

Spending Range	Classification
Under ₹100	Low-value orders
₹100–₹199	Budget-conscious orders
₹200–₹299	Mid-range orders
₹300–₹499	Premium orders
₹500+	High-value / large orders

The purpose is to understand how orders are distributed across different customer spending levels.

21. Rating Analysis

- Rating distribution from 1 to 5
- Number of orders associated with each rating
- Relationship between ratings and order volume
- High-rated restaurants with relatively low order volume

This analysis can help identify restaurants with strong customer satisfaction that may have opportunities to increase visibility and sales.

22. SQL Techniques Used

Area	Techniques
Data Cleaning	NULL validation, empty-string detection, duplicate detection and removal
Transformation	CTEs, CASE, aggregations, date functions, formatting
Joins	Fact and dimension joins, multi-table analytical queries
Window Functions	ROW_NUMBER(), LAG()
Analytics	COUNT(), SUM(), AVG(), ranking, percentage calculations
Data Modelling	Fact tables, dimension tables, foreign keys, Star Schema

23. Key Analytical Questions

Sales & Revenue

- How many total orders were placed?
- What is the total revenue?
- What is the average dish price?
- Which months generate the highest revenue?
- How does revenue change month over month?

Geography

- Which cities have the highest order volumes?
- Which states contribute the most revenue?
- Which cities have the highest average ratings?

Restaurants

- Which restaurants receive the most orders?
- Which restaurants generate the highest revenue?
- Which restaurants should be prioritized for promotion?
- Which restaurants have high ratings but relatively low order volume?

Food

- Which categories are most popular?

- Which dishes are ordered most frequently?
- Which cuisines perform best?
- Which cuisines combine high order volume with strong ratings?

Customer Behavior

- How are orders distributed across spending ranges?
- What does the customer rating distribution look like?

24. Project Results

- Loaded and profiled 1,97,401 raw records.
- Performed null-value checks.
- Performed empty-string validation.
- Detected duplicate records.
- Removed duplicate records.
- Created dimension tables.
- Created the central fact table.
- Implemented a Star Schema.
- Loaded dimension and fact data through SQL.
- Developed core business KPIs.
- Performed time-based analysis.
- Performed geographic analysis.
- Performed restaurant analysis.
- Performed food and cuisine analysis.
- Analyzed customer spending patterns.
- Analyzed rating distributions.
- Identified potential business opportunities.

25. Business Value

The project demonstrates how raw transactional data can be converted into actionable business information. The resulting analysis can help stakeholders monitor order performance, understand revenue trends, identify high-performing locations, evaluate restaurant performance, understand customer spending behavior, identify popular food categories and dishes, monitor customer satisfaction, and identify restaurants with growth potential.

The Star Schema also provides a structured foundation for future reporting and business-intelligence applications.

26. Skills Demonstrated

Skill Area	Demonstrated Capabilities
SQL	Querying, cleaning, aggregation, joins, CTEs, window functions, date analysis, business metrics

Skill Area	Demonstrated Capabilities
Data Analytics	KPI development, trend analysis, revenue analysis, geographic analysis, customer behavior analysis
Data Modelling	Star Schema, fact tables, dimension tables, relational modelling
Data Engineering Fundamentals	Data ingestion, cleaning, ETL, analytical data models
Business Analysis	Translating requirements into SQL analysis and identifying performance patterns

27. Conclusion

The Swiggy Sales Analysis project demonstrates a complete SQL-based data analytics workflow, beginning with raw food-delivery data and ending with structured business insights.

The project first established data quality through null, blank, and duplicate checks. The cleaned dataset was then transformed into a Star Schema consisting of date, location, restaurant, category, and dish dimensions connected to the central fact_swiggy_orders table.

After implementing the analytical model and ETL process, SQL was used to develop core KPIs and perform detailed business analysis across time, geography, restaurants, food categories, customer spending, and ratings.

Overall, the project demonstrates how SQL can be applied not only for querying data but also for data cleaning, dimensional modelling, ETL, KPI development, and business decision support.

28. Author

Adrij Das

Data Analyst Portfolio Project

Core Skills: SQL | Data Cleaning | Data Modelling | ETL | Business Analysis | Data Analytics